$$\sigma_n = \begin{pmatrix} n_z & (n_x - in_y) \\ (n_x + in_y) & -n_z \end{pmatrix}. \qquad (3.23)$$

What is this good for? Not much, until we find the eigenvectors and eigenvalues of σ_n. But once we do that, we will know the possible outcomes of a measurement along the direction of $\hat{n}$. And we will also be able to calculate probabilities for those outcomes. In other words, we will have a complete picture of spin measurements in three-dimensional space. That is pretty darn cool, if I say so myself.

3.7 REAPING THE RESULTS

We are now positioned to make some real calculations, something that should make your inner physicist jump for joy. Let's look at the special case where $\hat{n}$ lies in the *x*–*z* plane, which is the plane of this page. Since $\hat{n}$ is a unit vector, we can write

$$n_z = \cos\theta$$

$$n_x = \sin\theta$$

$$n_y = 0,$$

where θ is the angle between the *z* axis and the $\hat{n}$ axis. Plugging these values into Eq. 3.23, we can write

$$\sigma_n = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.$$

Exercise 3.3: Calculate the eigenvectors and eigenvalues of σ_n. *Hint:* Assume the eigenvector λ_1 has the form